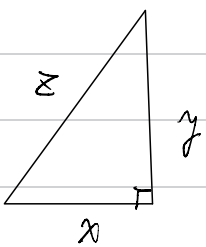
10/12/2026

Pythagorean triples



$x^2 + y^2 = z^2$, $x, y, z \in \mathbb{Z}$.

Eg. $(x, y, z) = (3, 4, 5)$, $(5, 12, 13)$.

Goal: Find all $x, y, z \in \mathbb{Z}$ satisfying

$$x^2 + y^2 = z^2.$$

Observation: If (x_0, y_0, z_0) is a solution, then

$(x_0 d, y_0 d, z_0 d)$ is also a solution, where $d \in \mathbb{Z}$.

Idea: Suppose $z \neq 0$ and $(x, y) = 1$.

Factor: $\underbrace{(x + yi)}_{\alpha}\underbrace{(x - yi)}_{\bar{\alpha}} = z^2$

If we can show

$$(x+y_i,\ x-y_i)=1,$$

then $x+y_i=\varepsilon\cdot\beta^2$, $\beta\in\mathbb{Z}[i]$

(ε: unit $=\pm 1,\pm i$)

Say $\beta=u+vi$, $u,v\in\mathbb{Z}$.

Then $\beta^2=(u+vi)^2=(u^2-v^2)+2uv\cdot i$

$$\varepsilon\beta^2=\begin{cases}(u^2-v^2)+2uv\cdot i & \varepsilon=1\\ -(u^2-v^2)-2uv\cdot i & \varepsilon=-1\\ -2uv+(u^2-v^2)i & \varepsilon=i\\ 2uv-(u^2-v^2)i & \varepsilon=-i\end{cases}$$

So $x + iy = \pm(u^2 - v^2) \pm 2uv \cdot i$

$$\text{or} = \pm 2uv \pm i(u^2 - v^2).$$

$$\Rightarrow (x, y) = (\pm(u^2 - v^2), \pm 2uv)$$

$$\text{or} = (\pm 2uv, \pm u^2 - v^2)$$

$$\Rightarrow z^2 = x^2 + y^2 = (u^2 - v^2)^2 + (2uv)^2$$

$$= (u^2 + v^2)^2$$

$$\Rightarrow z = \pm(u^2 + v^2).$$

Now we solve $x^2 + y^2 = z^2$ completely.

1. $z = 0$: $x = y = 0$.

2. $z \neq 0$: Then x, y are not both 0.

Let $d = (x, y)$. Then $d \mid x$, $d \mid y$ and so

$d^2 \mid x^2$ and $d^2 \mid y^2$. Thus, $d^2 \mid z^2$. Hence, $d \mid z$.

Write $x = x_1 d$, $y = y_1 d$, $z = z_1 d$. Then

$$x_1^2 + y_1^2 = z_1^2,$$

where $(x_1, y_1) = 1$.

If z_1 is even, $x_1^2 + y_1^2 \equiv 0 \pmod 4$. Since x_1^2, y_1^2

$\equiv 0, 1 \pmod 4$, we know $x_1^2, y_1^2 \equiv 0 \pmod 4$.

So $2 \mid x_1$, $2 \mid y_1$, contradicting $(x_1, y_1) = 1$.

Hence $2 \nmid z_1$. As a result, exactly one of x_1, y_1 is even.

Now factor

$$\underbrace{(x_1 + y_1 i)}_{\alpha}\underbrace{(x_1 - y_1 i)}_{\bar{\alpha}} = z_1^2.$$

Let $\pi \mid \gcd(\alpha, \bar{\alpha})$. Then $\pi \mid \alpha$, $\pi \mid \bar{\alpha}$. So

$$\pi \mid \alpha + \bar{\alpha} = 2x_1$$

$$\pi \mid \alpha - \bar{\alpha} = 2y_1 \cdot i \quad (i \text{ a unit}) \implies \pi \mid 2y_1.$$

$$\implies \pi \mid (2x_1, 2y_1) = 2\underbrace{(x_1, y_1)}_{2}$$

$\to$ gcd of x_1, y_1 in $\mathbb{Z}[i]$, but we showed that this is 1 when $(x_1, y_1) = 1$ in $\mathbb{Z}$.

So $\pi \mid 2$. Thus $N(\pi) = \pi\bar{\pi} = 2$. But

$$\pi \mid \alpha, \ \bar{\pi} \mid \bar{\alpha} \Rightarrow \underbrace{\pi\bar{\pi}}_{2} \mid \alpha\bar{\alpha} = z_1^2,$$

so z_1 is even, impossible. Hence, $(\alpha, \bar{\alpha}) = 1$.

Since $\alpha\bar{\alpha} = z_1^2$, $\exists$ unit ε and $\beta = u + iv \in \mathbb{Z}[i]$

s.t.

$$\alpha = \varepsilon \cdot \beta^2.$$

As sketched earlier,

$$x_1 = \pm(u^2 - v^2), \ y_1 = \pm 2uv, \ z_1 = \pm(u^2 + v^2)$$

or

$$x_1 = \pm 2uv, \ y_1 = \pm(u^2 - v^2), \ z_1 = \pm(u^2 + v^2).$$

Primitive solutions: $(u, v) = 1$, exactly one of which is even.

Thus

$$x = \pm(u^2 - v^2)d, \quad y = \pm 2uvd, \quad z = \pm(u^2 + v^2)d.$$

or

$$x = \pm 2uvd, \quad y = \pm(u^2 - v^2)d, \quad z = \pm(u^2 + v^2)d.$$

These are all solutions if we allow d to be 0.

Fermat's Last Theorem:

$$x^n + y^n = z^n \quad (n > 2)$$

has no solutions (x, y, z) with $x, y, z \neq 0$.

This was proved by Wiles and Wiles–Taylor (1995).

Thm (Primes in $\mathbb{Z}[i]$). Let $\pi \in \mathbb{Z}[i]$ be prime.

1. $\pi = \pm 1 \pm i \mid 2$.

2. $\pi = p \equiv 3 \pmod 4$ or one of its associates

3. $\pi = x + iy \mid p$ with $x, y \neq 0$, where $p \equiv 1 \pmod 4$.

In fact,

$$N(\pi) = \pi\overline{\pi} = x^2 + y^2 = p.$$

If π is such a prime divisor of p, there are

8 prime divisors of p in total!

$$\pm\pi, \pm i\pi, \pm\overline{\pi}, \pm i\overline{\pi}.$$

So there are exactly 8 ways of writing $p = x^2 + y^2$.